

Xuangeng CHU

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🌐 <https://xg-chu.site>

Research Interests

1. Human-centric 3D vision, digital human, shape / pose generation and reconstruction.
2. Novel view synthesis (NeRF / 3DGS).

Education

The University of Tokyo

Apr 2023 – now

Ph.D. Candidate in RCAST, University Fellowship, Supervisor: Prof. Tatsuya HARADA Tokyo, Japan

Peking University

Sep 2018 – Jun 2021

M.Eng. in Software Engineering, Supervisor: Prof. Yasha WANG Beijing, China

Tongji University

Sep 2014 – Jun 2018

B.Eng. in Computer Science Shanghai, China

Work Experience

Tencent, Research engineer

Mar 2021 – Oct 2022

Applied Research Center, work closely with Dr. Ying SHAN Shenzhen, China

- *Worked on category extensible object detection algorithm for videos.*
- *Worked on articulated model reconstruction algorithm for a virtual scene generating system.*

Internship Experience

Princeton University, Visiting student researcher

Jul 2024 – Aug 2024

Advised by Prof. Jia DENG New Jersey, USA

- *Worked on research of general one-shot 3D reconstruction.*

International Digital Economy Academy, Research intern

Dec 2022 - Nov 2023

Advised by Dr. Yu LI Shenzhen, China / Remote

- *Worked on research and development of human reconstruction and pose estimation.*

Microsoft Research Asia, Research intern

Jun 2020 - Feb 2021

Advised by Dr. Xiulian PENG Beijing, China

- *Worked on research of audio-visual speech separation problem (cocktail party problem).*

MEGVII Technology, Research intern

Jan 2019 - Jun 2020

Advised by Dr. Xiangyu ZHANG Beijing, China

- *Worked on research and development of object detection in crowded scenes.*

Publication

ARTalk: Speech-Driven 3D Head Animation via Autoregressive Model **Preprint**
Xuangeng Chu, Nabarun Goswami, Ziteng Cui, Hanqin Wang, Tatsuya Harada

We propose an efficient autoregressive framework capable of generating natural 3D facial motions with head poses in real time.

Luminance-GS: Adapting 3D Gaussian Splatting to Challenging Lighting Conditions with View-Adaptive Curve Adjustment **CVPR 2025**

Ziteng Cui, **Xuangeng Chu**, Tatsuya Harada

We introduce a novel approach to achieve high-quality novel view synthesis results under diverse and challenging lighting conditions using 3DGS.

Generalizable and Animatable Gaussian Head Avatar **NeurIPS 2024**

Xuangeng Chu, Tatsuya Harada

We propose the first generalizable 3DGS head avatar framework that achieves single forward reconstruction and real-time reenactment. The key idea of this work is lifting two sets of 3D Gaussian points from the input image. Project website and code: https://xg-chu.github.io/project_gaga.

GPAvatar: Generalizable and Precise Head Avatar from Image(s) **ICLR 2024**

Xuangeng Chu, Yu Li, Ailing Zeng, Tianyu Yang, Lijian Lin, Yunfei Liu, Tatsuya Harada

We propose a framework to reconstructs 3D head avatars from one or several images in a single forward pass. The key idea of this work is a dynamic point-based expression field and a attention-based fusion module. Project website and code: https://xg-chu.github.io/project_gpavatar.

Real-time High-resolution View Synthesis of Complex Scenes with Explicit 3D Visibility Reasoning **TVCG 2024**

Tiansong Zhou, Yebin Liu, **Xuangeng Chu**, Chengkun Cao, Changyin Zhou, Fei Yu, Yu Li

We propose a view synthesis method capable of real-time rendering of high-resolution novel-view images from sparse view inputs. Our method uses explicit 3D visibility reasoning as the core technique to address the occlusion problems of input views.

Accurate 3D Face Reconstruction with Facial Component Tokens **ICCV 2023**

Tianke Zhang, **Xuangeng Chu**, Yunfei Liu, Lijian Lin, Zhendong Yang, Zhengzhuo Xu, Chengkun Cao, Fei Yu, Changyin Zhou, Chun Yuan, Yu Li

We propose a framework for 3D face reconstruction from monocular images based on 3DMM and transformers. Our method uses separate tokens to improve the disentanglement of shape and expression for more accurate reconstruction.

Detection in Crowded Scenes: One Proposal, Multiple Predictions **CVPR 2020 Oral**

Xuangeng Chu*, Anlin Zheng*, Xiangyu Zhang*, Jian Sun

We propose a simple and almost cost-free method to improve the detection performance in crowded scens. The key idea of this work is to predict a set of instances from each proposal reagon instead of just one. Code is avaiable on: <https://github.com/xg-chu/CrowdDet>.

Services

Reviewer, TPAMI; CVPR 2022, 2023; ACMMM 2024; NeurIPS 2024; ICLR 2025.